

Microarray

probe to symbol · log2
mean-collapse duplicates

RNA-seq counts (TCGA)

TPM · log2(TPM+1)
mean-collapse duplicates

RNA-seq FPKM

log2(FPKM+1)
sum additive sub-features

Convert to Ensembl gene IDs (cached map)

log2, gene-indexed expression matrix

Subtype callers

CMS, PDS use a
log2 symbol matrix

ssGSEA scoring · 70% coverage gate

Signature scores

Up · Down · Composite (+ 6 reference sets)

Deterministic under one global seed; only GSEA tie-breaking is stochastic.